

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms equation to find the centre of mass.	AO1.1a	M1	$\bar{x} = \frac{3.7 \times 20 + 2 \times 22.5}{42.5} = 2.8$
	Obtains correct answer.	AO1.1b	A1	
(b)	Forms an equation using moments.	AO1.1a	M1	Moments about C: $42.5 \times 9.8 \times 2.3 = 2.5R_D$ $R_D = \frac{957.95}{2.5}$ $= 383.18 \text{ N}$ $= 380 \text{ N (2sf)}$
	Obtains correct moment equation, follow through 'their' centre of mass provided both M1 marks have been awarded	AO1.1b	A1F	
	Obtains one correct reaction force for 'their' equation. Allow more than 2 sf accuracy FT provided both M1 marks awarded	AO1.1b	A1F	
	Obtains both reaction forces correctly to 2 sf accuracy, follow through 'their' reaction force, provided both M1 marks have been awarded.	AO1.1b	A1F	
				$R_C + R_D = 42.5 \times 9.8$ $R_C = 416.5 - 383.18$ $= 33.32 \text{ N}$ $= 33 \text{ N (2sf)}$
				Total 6 marks

Q2.

(a) Symmetry

E1

1

(b) Moments about AB:

$$300\sigma.15 + 100\sigma.5 + 300\sigma.15 = 700\sigma. x$$

(condone lack of σ .)

M1 needs correct total marks

$$x = \frac{9500}{700}$$

M1A1

$$= \frac{95}{7} \text{ or } 13.6 \text{ cm}$$

A1

3

- (c) Distance from HG is 16.4 cm

B1

$$\tan \theta = \frac{15}{16.42857}$$

Seeing both 15, 16.4 and tan

M1

$$= 0.913043$$

$$\theta = 42.3974^\circ$$

A1

$$\theta = 42^\circ$$

[48° probably B1, M1]

NB $\frac{13.6}{15}$ etc $\Rightarrow 42^\circ$ no marks

A1

4

[8]

Q3.

- (a) (i) Moments about AB :

$$1.6 \times 4 + 0.4 \times 8 = 2 \times x$$

M1 for 2 terms correct

M1A1

$$x = 4.8$$

Distance is 4.8 cm

A1

3

- (ii) Moments about AD :

$$1.6 \times 6 + 0.4 \times 12 = 2 \times y$$

M1 for 2 terms correct

M1A1

$$y = 7.2$$

Distance is 7.2 cm

SC2 + SC2 for (a)(i) and (a)(ii) reversed

A1

3

(b) Moments about A:

$$1.6g \times 6 + 0.4g \times 12 = 12 \times T_B$$

M1 for 1 side of equation

Or using above: moments about A

$$12 \times T_B = 7.2 \times 2g \quad (\text{ft for M marks})$$

M1A1

$$T_B = 1.2g = 11.8 \text{ N}$$

A1

Resolve vertically: $T_A + T_B = 2g$

M1

$$T_A = 0.8g = 7.84 \text{ N}$$

1.2 and 0.8 is zero marks

If 11.8 and 7.8 as final answer, must lose 1 mark somewhere

A1

5

[11]

Q4.

(a) Moments about line AD: $5 \times 30 + 4 \times 10 = 9 \times \bar{x}$

M1: 2 of 3 terms correct

M1A1

$$\bar{x} = \frac{190}{9} = 21.1 \text{ cm}$$

A1

3

(b) Moments about line AB: $5 \times 15 + 4 \times 25 = 9 \times \bar{y}$

M1: 2 of 3 terms correct

M1A1

$$\bar{y} = \frac{175}{9}$$

$$\bar{y} = 19.4 \text{ cm}$$

If moments about DC; 10.6 found SC2

A1

3

(c) $\tan \theta = \frac{80}{175}$ or $\frac{8.9}{19.4}$

M1: use of tan

M1

A1: use of 8.9 or 80 (30 – (a))

A1

$= 0.4571$

Or 0.45876

A1

Angle is $\tan^{-1} 0.4571 = 24.6^\circ$

$65.4^\circ \Rightarrow$ M1A1 only

A1

4

(d) Moments about the line *PR*: (or *AD* or *BC*)

M1

$30m = 4 \times 20$ or $9 \times \frac{80}{9}$

A1

$m = \frac{8}{3}$

A1

3

(e) Centre of mass is at middle of lamina

E1

1

[14]

Q5.

(a) Symmetry

Only accept 'symmetry'

E1

1

(b) Moments about *B*:

$0.4 \times 4 + 0.1 \times 8 = 0.5 \times \bar{x}$

M1 3 terms, 2 correct

M1A1

$$\begin{aligned}\bar{x} &= \frac{2.4}{0.5} \\ &= 4.8 \text{ cm}\end{aligned}$$

A1

3

[4]

Q6.

- (a) Taking moments about AD :

$$8 \times 10 + 2 \times 15 = 10 \bar{x}$$

*M1 for moments and 1 term on left
correct and 1 term on right*

M1A1

$$\begin{aligned}\bar{x} &= \frac{110}{10} \\ &= 11 \text{ cm}\end{aligned}$$

A1

3

- (b) 5 cm

B1

1

- (c) $(\tan)\theta = \frac{1}{5}$ ie $\frac{(a) - 10}{(b)}$

From areas; $\frac{1.4}{5} \Rightarrow \theta = 15.6$ or 15.7

M1

$$= 0.2$$

A1ft

Angle is $\tan^{-1}(0.2)$

M1

$$= 11.3^\circ$$

A1ft

4

- (d) Centre of mass is at middle of lamina

E1

1

Q7.

- (a) Symmetry of the lamina about BD

E1

1

- (b) Taking moments about AC :

$$40\rho \times 20$$

Condone lack of ρ

M1

$$= 80\rho \bar{x}$$

A1

$$\bar{x} = 10 \text{ cm}$$

Answer 10 NMS = full marks

A1

3

- (c) $\tan \theta = \frac{10}{20}$

M1

Angle is $26.565\dots$

A1

$\approx 27^\circ$

AWRT 27

Angle is 63°

A1

3

[7]

Q8.

- (a) $M(AB)$

$$(0.8 \times 1.0)\rho \times 0.5 + (2.0 \times 0.2)\rho \times 2.0 = 1.2\rho \bar{x}$$

M1 all terms

-1 EE from A2

M1A2

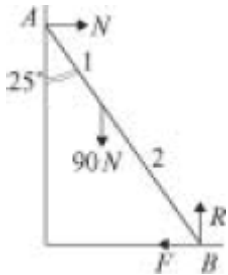
$$\bar{x} = 1, M \text{ on } CH$$

reference to position of M on CH needed

A1

4

(b)



(i) $M(B) : 90 \times 2 \times \sin 25^\circ = N \times 3 \times \cos 25^\circ$

Must attempt 'components' for M1

M1A1

$$N = 60 \tan 25^\circ$$

A1

3

(ii) $F = N \quad R = W = 90$

B1

$$F \leq \mu R$$

$$60 \tan 25^\circ \leq \mu \times 90$$

*Condone equality for M1, correct
Substitution for A1*

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Correct use of inequality throughout

A1

$$u \geq \frac{2}{3} \tan 25^\circ$$

4

[11]

Q9.

(a) $(10 \times 40)\rho \times 5 + (10 \times 60)\rho \times 40$

M1

$$= (10 \times 40 + 10 \times 60)\rho \checkmark$$

M1A1

$$\bar{y} = 26 \text{ cm}$$

A1

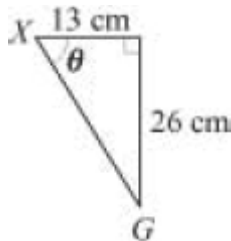
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- (b) Symmetry of shape

B1

1

- (c)



Attempting subtraction leading to 13 cm

M1

$$\tan \theta = \frac{26}{13}$$

Or inverted, must see 26

M1

Or inverted

A1

$$\theta = 63^\circ$$

$$(63.4)$$

Accept 117°

A1

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4

[9]

Q10.

- (a) Symmetry of the lamina about PQ

Accept 'mirror line'

E1

1

- (b) Taking moments about AB :

$$600\rho \times 15 + 100\rho \times 35$$

Condone lack of ρ

M1A1

$$= 700\rho \bar{x}$$

A1

$$\bar{x} = 17.857 = 17.9 \text{ cm}$$

SC3 17.8

A1

4

(c) $\tan \theta = \frac{10}{17.857}$

M1 for use of $\tan \theta$

M1A1

$$= 0.56$$

Angle is 29.2488...

M1

$$= 29^\circ$$

A1

4

[9]

Q11.

- (a) Because the lamina is symmetrical.
- Correct explanation*

B1

1

(b) $\bar{y} = \frac{250 \times 2.5 + 150 \times 7.5}{250 + 150}$

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Moment equation with appropriate number of terms

M1

Correct numerator

A1

$$= \frac{1750}{400}$$

correct denominator

A1

$$= 4.375$$

AG
Correct value from correct working

A1

4

(c) $\tan \alpha = \frac{10 - 4.375}{25} = \frac{5.625}{25}$

Use of tan.

M1

Subtracting from 10

M1

Correct expression

A1

$\alpha = 12.7^\circ$

Correct angle

A1

4

- (d) When it has been assumed that the centre of mass of each of the rectangles used is at its centre.

OR

Relating area to mass.

Correct explanation

B1

1

[10]

Q12.

- (a) (i) Use of $I = \frac{\int xy dx}{\int y dx}$
[$\rho = 1$ allowed throughout]
Stated or used

M1

$$\int y dx = \text{semi circle area} = \frac{1}{2} \pi r^2$$

A1

Use of diagram / by symmetry, $\bar{x} = 21$
Explains need for $2 \times$ Integral

A1

$$\frac{1}{2} \pi r^2 \bar{x} = \int_0^r 2x \sqrt{r^2 - x^2} dx$$

Use of $y = \sqrt{r^2 - x^2}$ and rearrangement

Alternative

Mass of elemental strip $= 2y \delta x \rho$

[Allow $\rho = 1$ throughout]

Elemental strip identified

Moment of elemental strip $= 2y \delta x \rho x$

(M1)

Total moment $= \rho \int_0^r 2xy \, dx$
Integral formed

(A1)

Use of $\sum(mx) = (\sum m)\bar{x}$

Gives $\rho \int_0^r 2xy \, dy = \frac{1}{2} \rho \pi r^2 \bar{x}$
Equation formed

(M1)

$$\Rightarrow \int_0^r 2x \sqrt{r^2 - x^2} \, dx = \frac{1}{2} \pi r^2 \bar{x}$$

Use of $\sqrt{r^2 - x^2} = y$ and $\frac{1}{2} \pi r^2$

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 (A1)

(ii) $\int_0^r 2x \sqrt{r^2 - x^2} \, dx = \left[-\frac{2}{3} (r^2 - x^2)^{\frac{3}{2}} \right]_0^r$

Attempt to integrate – inspection or substitution

M1

*Integrated correctly and limits substituted –
 condone sign error*

A1

$$= (0) - \left(-\frac{2}{3} r^3 \right)$$

$$= \frac{2}{3} r^3$$

$$\therefore \frac{1}{2} \pi r^2 \bar{x} = \frac{2}{3} r^3$$

$$\bar{x} = \frac{4r}{3\pi}$$

Printed answer

A1

3

(b) (i) $\frac{2}{3}(1.2) = 0.8 \text{ m}$

B1

1

(ii) $1.2 + \frac{4(0.5)}{3\pi} = 1.41 \text{ m}$

B1

1

(iii) Using $(\sum m) \bar{x} = \sum (mx)$

$$\left[\frac{1}{2}(1)(1.2) \right] [0.8] + \left[\frac{1}{2} \pi (0.5)^2 \right] [1.41]$$

$$= \left[\frac{1}{2}(1)(1.2) + \frac{1}{2} \pi (0.5)^2 \right] \bar{x}$$

Form equation – Follow through (b)(i)

M1

One term correct

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A1ft

$$0.48 + 0.553... = 0.992... \bar{x}$$

All terms correct

A1ft

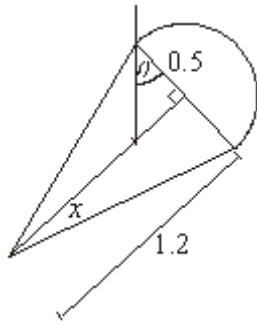
$$\therefore \bar{x} = 1.04 \text{ m}$$

CAO

A1

4

(c)



$$\tan \theta = \frac{(1.2 - 1.04)}{0.5} = 0.32$$

tan q seen

M1

$$\text{Ratio correct structure} = \frac{(1.2 - \bar{x})}{0.5} = 0.32$$

A1

$$\theta = 18^\circ$$

ft ratio error – correct use of $\tan^{-1}$ (ratio)

A1ft

3

[16]

Q13.

$$(a) \quad M(A) \quad 4Mg \times \frac{3d}{2} + Mg \times 2d = 5Mg \times \bar{y}$$

M1 3 terms (M1 A0 if areas used)

Condone ratio methods for weights.

–1 each term wrong

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M1A2

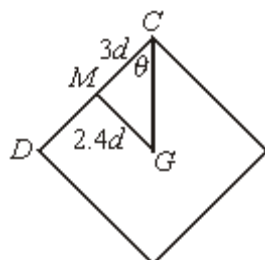
$$\bar{y} = 1.6d$$

$$M(AB) \quad 4Mg \times \frac{3d}{2} + Mg \times 2d = 5Mg \times \bar{y}$$

A1

4

(b)



$$\tan \theta = \frac{GM}{CM}$$

Full method for an acute angle involving wallet

M1

$$= \frac{2.4d}{3d}$$

A1A0 for inverse

A1A1

$$\theta = 38.7^\circ$$

ft slip in subtraction

A1F

4

[8]

Q14.

(a) $3(2M) + 3(3M) = 15M$

B1

1

(b) (i) From BC, $\sum Mx = (\sum M)\bar{x}$
Attempt to use (one term correct)

M1

$$2M(0.6) + 3M(0.15) + 3M(0.15) + 3M(0.45) + 2M(0.3) = 15M\bar{x}$$

2 terms

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A1f.t.

All correct

A1f.t.

$$\bar{x} = 0.27 \text{ metres}$$

AG, ft incorrect part (a)

A1

4

(ii) From CE $\sum My = (\sum M)\bar{y}$

$$2M(0.1) + 2M(0.1) + 2M(0.1) + 3M(0.2)$$

$$= 15M\bar{y}$$

M1 – one term correct

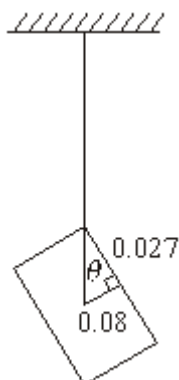
M1A1✓

$$\bar{y} = 0.08 \text{ metres}$$

A1

3

(c)



$$\tan \theta = \frac{\bar{y}}{\bar{x}}$$

Application

M1

$$= \frac{8}{27} \text{ or } 0.0296$$

$\bar{x}$ and their $\bar{y}$

A1ft

$$\theta \approx 16.5^\circ$$

CAO

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A1

3

[11]

Q15.

(a) (i) Distance = $4 + \frac{1}{3} \times 6 = 6$

M1A1

2

(ii)	Shape	Mass	Dist from AB	Mass × Dist
	ABEC'	24ρ	2	48ρ
	C'ED	36ρ	6	216ρ
	ABED	60ρ	$\bar{X}$	$60\rho \bar{X}$
		$\therefore 60\rho \bar{X}$	$= 264\rho$	
		$\bar{X}$	$= 4.4$	

no penalty if ρ omitted

M1

CAO; AG

A1

2

(iii)	Shape	Mass	Dist from AD	Mass $\times$ Dist
	$ABEC'$	24ρ	3	72ρ
	$C'ED$	36ρ	2	72ρ
	$ABED$	60ρ	$\bar{Y}$	$60\rho \bar{Y}$
		$\therefore 60\rho \bar{Y}$	$= 144 \rho$	
		$\bar{Y}$	$= 2.4$	

A1

2

(b) Moments about $B : K = 0.044 \times 2 = 0.088$

M1A0 for 8.8

M1A1

2

[8]

Q16.

areas	distance from N
small $\pi(1)^2$	2
large $\pi(2)^2$	3
earring $\pi(2)^2 - \pi(1)^2$	$\bar{x}$

attempt to find area of one circle (evidence of πr^2)

M1

using $\sum (mx) = (\sum m)\bar{x}$

$3\pi\bar{x}$ or (difference of their areas) $\bar{x}$

M1

$$2(4\pi) - 3(\pi) = 3\pi\bar{x}$$

one other moment evident

M1

$$5\pi = 3\pi\bar{x}$$

A1

$$\bar{x} = \frac{5}{3} \text{ or } 1\frac{2}{3} \text{ or } 1.67\text{cm}$$

fully correct

A1

5

[5]

Q17.

Moment about A,

$$0.4(0.25) + 0.95(1) = 1.25\bar{x}$$

Attempt to use $(\sum m)\bar{x} = \sum(mx)$
 A1 one term correct
 A1 all correct

M1 A1 A1

$$\bar{x} = 0.84 \text{ m}$$

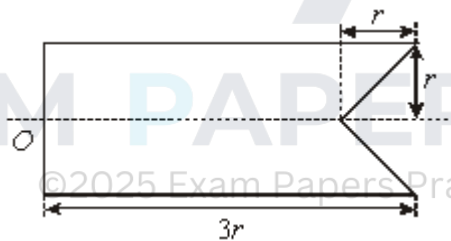
CAO

A1

[4]

Q18.

(a)



	Weight	Dist from O
Cylinder	$3\pi r^3 \rho$	$\frac{3r}{2}$
Cone	$\frac{1}{3}\pi r^3 \rho$	$2r + \frac{3r}{4} = \frac{11r}{4}$
New body	$\frac{8}{3}\pi r^3 \rho$	$\bar{x}$

3 shapes

M1

$-1EE$

A2

$$\frac{8}{3} \pi r^3 \rho \bar{x} + \frac{1}{3} \pi r^3 \rho \times \frac{11r}{4} = 3 \pi r^3 \rho \times \frac{3r}{2}$$

correct combination

M1

at least 2 weight $\times$ dist correct

A1

$$\frac{8\bar{x}}{3} = \frac{9r}{2} - \frac{11r}{12}$$

$$\Rightarrow 32\bar{x} = 54r - 11r$$

$$\Rightarrow \bar{x} = \frac{43r}{32}$$

A1

6

(b)



Body will topple when $\alpha > \tan^{-1} \frac{r}{\bar{x}}$

M1

$$\alpha > \frac{\tan^{-1} \frac{r}{\bar{x}}}{\frac{43r}{32}}$$

A1f.t.

$$> \tan^{-1} \frac{32}{43}$$

$$> 36.7^\circ$$

A1f.t.

3

[9]

Q19.

- (a) AP is a line of symmetry
must refer to symmetry

E1

1

(b)

	Area	Dist of C of M from CD
ABCD	160	4
PQRD	96	14
	256	$\bar{x}$

2 correct areas
seen or implied use

B1

Using $\Sigma(mx) = \Sigma m(\bar{x})$

then $160(4) + 96(14) = 256$

M1 correct distance paired with correct areas

M1A1

$$\bar{x} = 7.75$$

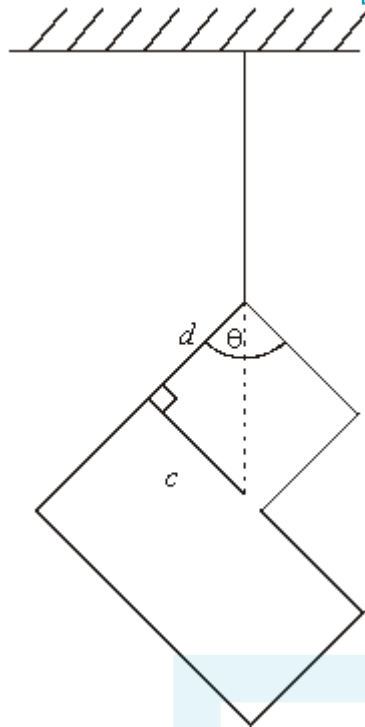
ft one slip in areas or distance

A1f.t.

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4

(c)



$$\tan \theta = \frac{c}{d}$$

Intention to apply principle

M1

$$c = 7.75$$

c correct ft part (b) ($c = \bar{x}$)

A1f.t.

$$d = 12.25$$

d correct ft part (b) ($d = 20 - \bar{x}$) or their $\bar{y}$

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A1f.t.

$$\Rightarrow \theta = 32^\circ$$

AWRT

A1

4

[9]

Q20.

(a) $\text{Area} = 16 + 4 = 20 \text{ cm}^2$

Correct area of lamina

B1

$$\bar{x} = \frac{16 \times 2 + 4 \times 6}{20}$$

Three term moment equation

M1

Correct equation

A1

ag Correct distance from correct working

A1

4

(b)
$$\bar{y} = \frac{16 \times 2 + 4 \times 3.5}{20}$$

Three term moment equation

M1

Correct equation

A1

= 2.3 cm

Correct distance

A1

3

(c)
$$\tan \alpha = \frac{2.3}{2.8}$$

Use of tan

M1

Correct equation

A1

$\alpha = 39.4^\circ$

Correct angle

A1

3

[10]

Q21.

- (a) If $\bar{x}$ is the distance of the centre of mass from the base
Moments about plane of the base

M1

$$\pi r^2 \cdot 2r \cdot r \cdot \rho + \frac{1}{3} \pi r^2 H (2r + \frac{H}{4}) \cdot 3\rho$$

M1A1

$$= (\pi r^2 \cdot 2r\rho + \frac{1}{3} \pi r^2 H \cdot 3\rho) \bar{x}$$

A1

$$2r^2 + 2Hr + \frac{H^2}{4} = (2r + H) \bar{x}$$

Dep both M1

m1

$$\bar{x} = \frac{8r^2 + 8Hr + H^2}{4(2r + H)}$$

A1

6

(b) If on the point of toppling

$$\tan \theta = \frac{1}{5} = \frac{r}{\bar{x}}$$

$$\therefore 5r = \frac{8r^2 + 8Hr + H^2}{4(2r + H)}$$

for $\frac{r}{5}$ or $5r =$ result in (a)

M1

A1

$$40r^2 + 20rH = 8r^2 + 8rH + H^2$$

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M1

$$32r^2 + 12rH - H^2 = 0$$

$$H^2 - 12rH - 32r^2 = 0$$

A1

$$H = \frac{12 \pm \sqrt{144 + 128}}{2} r$$

$$= (6 \pm 2\sqrt{17})r$$

$$= (6 + 2\sqrt{17})r$$

A1

5

[11]

Q22.

(a) $8\bar{x} = 1 \times 6 + 2.4 \times 2$

Three term moment equation

M1

Correct equation

A1

$$\bar{x} = \frac{10.8}{8} = 1.35$$

ag *Correct value form correct working*

A1

3

(b) $\tan \alpha = \frac{0.4}{1.05}$

Use of tan or sin/cos plus finding hypotenuse

M1

Use of 0.4

A1

Correct trig expression

A1

$$\alpha = 20.9^\circ$$

Correct angle

A1

4

[7]

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Q23.

(a) $24\bar{x} = 16 \times 2 + 8 \times 6$

Forms moment equation based on two or three rectangles

M1

$$\bar{x} = \frac{80}{24} = \frac{10}{3}$$

Correct equation

Rearranged for x by dividing by total area

Correct final answer from correct working

A1/M1/A1

4

(b) $24\bar{y} = 16 \times 2 + 8 \times 1$

Forms moment equation based on two or three rectangles

M1

$$\bar{y} = \frac{40}{24} = \frac{5}{3}$$

Correct equation

Correct final answer from correct working

A1/A1

3

(c) $\tan \alpha = \frac{\frac{5}{3}}{\frac{10}{3}} = \frac{1}{2}$

Uses tan

Uses both distances from above

M1/A1

$$\alpha = 26.6^\circ$$

Correct equation

Correct angle

M1/A1

4

[11]

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Examiner reports

Q2.

In part (a) a number of students did not state that the reason the centre of mass was 15 cm from AH was as a result of the lamina being symmetrical. Part (b) of this question was usually answered well. A significant proportion of students, in part (c), did not find the distance of the centre of mass from HC . Others found the angle between HG and the vertical rather than the horizontal. A significant proportion of those who did obtain the angle to be 42.39° forgot to give the angle to the nearest degree, as required.

Q3.

Part (a) of this question was usually answered well. A significant proportion of candidates, in part (b), tried to use angles and found the distance of the centre of mass to one of the points, often A . Some forgot to involve g in their calculations.

Q4.

Many candidates answered parts (a), (b), (d) and (e) of this question well. The common error in parts (a) and (b) was in not using the position of the centre of mass of the triangular lamina correctly. Some considered areas rather than masses. In part (c), many candidates found the angle with the vertical, rather than with the horizontal as required. In part (e), a common error was in stating that the mass acted at the centre of mass, rather than at the centre of the lamina.

Q5.

In part (a), a number of candidates tried to explain why the centre of mass is on the line AB by discussing uniform laminas; the required answer was based on symmetry. Part (b) was completed well, although a number of candidates when taking moments about B used 6 cm as the distance of the centre of mass of the smaller circular lamina from B , rather than 8 cm. Some candidates considered areas rather than masses, assuming that the two laminas had the same density. On this occasion, this method was condoned.

Q6.

Unfortunately, many candidates answered the question which they wished to have been asked, namely that the two laminas were of equal density and their masses depended on their areas.

Parts (b) and (c) were normally answered correctly, often on a follow through. A significant number of candidates were unable to explain clearly how they had used the fact that each lamina was uniform.

Q7.

Candidates found this question straightforward and answered it well

Q8.

Part (a) was done well although not all candidates related their coordinates of the centre of mass to the line CH . Part (b) was mostly done well but those taking moments about a

point other than the .foot. of the lamina sometimes omitted a force. Candidates knew the principle of mechanics required in answering part (c). However, weaknesses in handling inequalities spoiled many otherwise good solutions.

Q9.

Parts (a) and (b) were done very well. In part (c) those who correctly identified the relative positions of X and the centre of mass of the lamina usually completed the question successfully.

Q10.

In part (a), many candidates used the word “symmetrical” in their explanations but others seemed to feel that an equal area on each side of PQ was adequate. In part (b), most recognised the need to take moments, and usually did so about the line AB . The common error was to use 5cm for the distance of the centre of mass of area $EFGH$, being the distance from EF , rather than 35cm, the distance from the line AB . In part (c), many candidates did not give the answer to the nearest degree.

Q11.

Part (a) of this was generally done quite well, although some candidates did use a moment equation as their explanation. Part (b), was either done very well or very badly. Almost all of the candidates who were able to make a good start completed this part without difficulty. In part (c), finding the angle correctly caused more difficulties. The main stumbling block was that candidates did not subtract the answer to part (b) from 10 before using trigonometry to find the angle. There were many good answers to part (d).

Q12.

Candidate performed better than expected here. In part (a)(i) two methods were credited. The standard ‘book’ method using elemental strips was rarely seen, although this is a useful skill for candidates to develop. Candidates who quoted the standard formula could gain full credit but the final A1 was for explaining from where the ‘2’ had originated. Part (a)(ii) proved more problematic – the integration being beyond many candidates. The best solutions used inspection of substitution with a variety of successful substitutions used including use of trigonometry. Many candidates scored full marks on part (b)(i) and part (b)(ii) although not all candidates were aware of the significance of part a) or the standard result for a triangle. Careless errors were seen in part (b)(iii) with the area of the semicircle being stated as $\pi/2$ or $\pi/4$. The best solutions started with a table giving clear values for areas and distances. Almost all of the candidates scored either 2 or 3 marks in part (c) and this is clearly well understood by candidates.

Q13.

A popular question. Part (a) was done well by most, with the most frequent error being the assumption that the masses of the pass and its wallet depended on their areas. Part (b) was often answered correctly, with only occasional errors in trigonometry and accuracy. Diagrams proved helpful to many candidates in selecting the appropriate ratio for their solution.

Q14.

Part (a) was correctly answered by almost all candidates. Many candidates showed

their competence with using $\sum (Mx) = (\sum M)x$ in part (b) and produced excellent solutions. Problems arose when candidates combined rods together for more efficient working and then used the wrong centre of mass for that combination. A few weak candidates thought they had to use the area of the rectangle. Part (c) was done better than in the past with the vast majority picking up marks for using $\tan \theta$ with their relevant distances.

Q15.

Part (a)(i) proved straightforward for the majority of candidates. Errors in parts (a)(ii) and (iii) usually resulted from not taking the double thickness of the triangular section into consideration. In part (b) the idea of the lamina being held in equilibrium by the action of a couple proved difficult for many.

Q16.

This question proved more demanding than expected. Several candidates did not know

the formula for the area of a circle, with $2\pi r$, $2\pi r^2$ and $\frac{\pi r^2}{2}$ often being seen. Many candidates then made one of two mistakes: either adding the moments of the individual areas together; or using 5π as the total of the remaining area. A few candidates tried to argue with ratios but were unsuccessful and scored very few marks. Fully correct solutions were therefore not that common.

Q18.

This was a situation where the final body was obtained by subtracting the cone from the cylinder, and this caused candidates difficulties in drawing up an appropriate “moments equation”. With the answer to the first part having been given, the solution to the final “toppling” problem was standard of its type. Candidates should be aware that they are not penalised here if they have not satisfactorily answered the first part of such a question, but go on to use the information given to pick up later marks.

Q19.

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This was another question that resulted in a sound response. The first mark was given for mentioning the symmetry of the system – simply commenting on equal areas was not sufficient. Marks were sometimes dropped here because candidates failed to

realise that areas were required when using $\sum (mx) = (\sum m)x$.

In the final part, some failed to identify the correct right-angled triangle or slipped up with the basic trigonometry. Candidates should be aware that the triangle will only involve using a tangent function.

Q20.

The candidates found this question more difficult. There were a number of candidates who made very little progress and also a number who were helped by the answer given in part (a). In part (c), some candidates found the distance of the centre of mass from the wrong side. If the candidates had the correct results in parts (a) and (b), they were generally able to complete part (c). Those who had incorrect distances usually had further problems when they used the wrong distances in part (c). A few candidates used an inappropriate trig function to find the angle.

Q21.

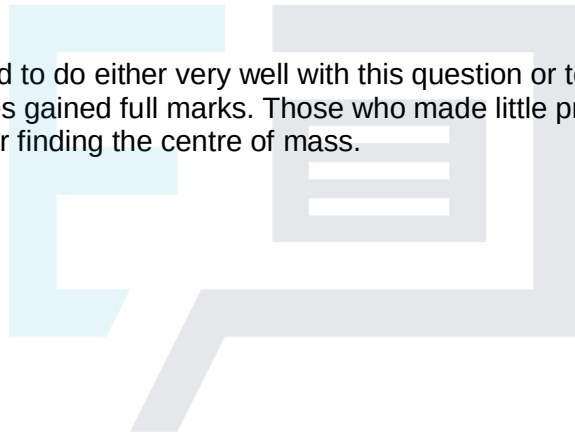
Better candidates made good progress with this question. In part (a) problems encountered by other candidates included finding the volume of a cone, in which the third in the formula $\frac{1}{3}\pi r^2 h$ was omitted or was incorrect; the mass of the cone, in which the density 3ρ was not used; and finding the position of the centre of mass of the cone. Part (b) was answered well by many candidates.

Q22.

Part (a) was done well by many candidates. There tended to be either complete solutions from those who knew how to tackle the problem, or virtually no working at all from those who did not know how to proceed. The printed answer seemed to have helped some candidates. The responses to part (b) were not so good. Many candidates used $\tan$ and the distance of 0.4 but did not put these together in the right way. Some candidates found the angle between the pole and the horizontal instead of the angle between the pole and the vertical.

Q23.

Candidates tended to do either very well with this question or to make very little progress. Several candidates gained full marks. Those who made little progress did not seem to have a strategy for finding the centre of mass.



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